

INFRATHRONE (COREOPS TRACK)

WEEK 1 ASSIGNMENT PACK (Linux + Kubernetes Fundamentals)

Theme: Building the Foundation of Outage Handling

This assignment is designed to push you slightly outside your comfort zone while keeping everything achievable.

You must submit screenshots, notes, and your debugging explanation for each task.

The goal is not perfection; the goal is understanding.



Mini Outage Simulation (Capstone)

Objective:

Bring Week-1 concepts together in a realistic sequence.

Scenario:

A pod is running, but the application is not responding. Debug the entire flow.

Steps:

1. Deploy: **kubectl run broken --image=nginx --command -- /bin/sh -c "sleep 3600"**
2. Add a liveness probe that intentionally fails: **kubectl edit pod broken**

Add:

```
livenessProbe:
  httpGet:
    path: /
    port: 8080
  initialDelaySeconds: 5
  periodSeconds: 5
```

3. Observe:
 - a. **kubectl describe pod broken**
 - b. **kubectl logs broken**
 - c. **kubectl get events**

4. Fix by:

- a. Correcting port**
- b. Or removing the probe**

Submission:

- Snapshot of CrashLoopBackOff
- Root cause explanation
- One paragraph: “What is the InfraThrone debugging pattern I used here?”

